

BILECIK, Turkey

WMO: 171200

Lat: 40.1502N

Lon: 29.9684E

Elev: 539

StdP: 95.02

Time Zone: 3.00 (E03)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-9.1	-15.2	1.1	-10.2	-12.8	1.3	-7.6	9.5	1.2	8.3	2.3	1.5	40	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	34.6	19.2	32.9	18.8	31.1	18.2	21.5	30.2	20.6	29.1	19.8	27.9	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.6	25.5	17.9	13.7	24.7	16.7	12.7	24.1	65.1	30.1	61.7	29.3	58.8	28.0	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.3	6.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	0.2	1.6	7.3	11.0	15.7	19.7	23.4	23.0	17.8	13.7	8.0	2.5
	DBStd	8.72	4.91	4.27	4.53	3.95	3.19	3.27	2.77	3.03	3.19	3.91	4.28	3.98
	HDD10.0	1012	303	236	105	32	1	0	0	0	0	13	86	236
	HDD18.3	2708	562	468	341	222	93	21	1	4	47	148	310	492
	CDD10.0	1764	1	1	22	63	178	292	416	402	233	128	26	2
	CDD18.3	419	0	0	0	2	12	63	159	148	30	4	1	0
	CDH23.3	5116	0	0	2	43	176	801	1873	1656	481	83	1	0
CDH26.7	2094	0	0	0	5	29	277	878	748	139	19	0	0	
Wind	WSAvg	2.3	1.8	2.3	2.4	2.5	2.3	2.8	3.0	3.2	2.6	1.9	1.5	1.7
Precipitation	PrecAvg	446	50	39	43	40	50	37	20	14	20	37	38	58
	PrecMax	625	120	135	92	75	118	131	106	68	70	112	88	158
	PrecMin	321	10	10	5	8	12	3	0	0	0	2	6	2
	PrecStd	78	32	22	22	17	31	27	24	17	19	25	20	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	15.7	22.2	27.2	29.2	34.3	37.2	35.9	32.0	29.1	22.2	14.7
		MCWB	9.1	8.8	11.2	14.5	17.4	20.1	19.7	19.0	18.0	18.2	13.1	9.3
	2%	DB	10.9	13.2	20.1	24.3	27.2	31.6	35.2	34.1	30.1	25.5	19.4	12.0
		MCWB	7.0	7.8	10.6	14.0	15.9	18.3	19.2	19.4	17.0	14.9	12.1	8.2
	5%	DB	8.9	11.1	18.0	21.9	25.2	29.9	33.1	32.8	28.2	23.0	17.2	9.8
		MCWB	5.8	6.9	9.9	12.6	15.3	17.5	18.8	18.8	16.5	14.0	11.0	6.6
	10%	DB	7.1	9.0	15.7	19.2	23.2	28.0	31.2	31.0	26.2	20.7	15.1	8.0
		MCWB	5.2	5.8	9.1	11.6	14.5	16.7	18.3	18.3	16.0	13.4	9.9	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.3	12.5	16.0	19.2	22.1	22.7	22.9	20.2	19.2	13.7	9.9
		MCDB	13.5	14.1	20.3	24.1	26.2	31.5	32.1	31.5	28.6	28.7	20.7	14.0
	2%	WB	7.2	8.5	11.3	14.4	17.7	20.6	21.3	21.7	18.7	16.4	12.5	8.4
		MCDB	9.8	12.0	18.6	22.9	24.1	28.2	30.4	30.2	26.9	21.6	18.2	11.4
	5%	WB	6.2	7.2	10.3	13.1	16.4	19.2	20.5	20.6	17.6	15.4	11.5	7.2
		MCDB	8.5	10.8	16.5	20.6	22.3	26.5	29.5	28.7	25.5	19.9	16.6	9.3
	10%	WB	5.0	5.7	9.4	12.0	15.3	18.0	19.7	19.7	16.6	14.5	10.4	5.9
		MCDB	7.0	8.6	15.0	18.5	21.4	25.4	28.2	27.6	24.5	18.7	14.5	7.8
Mean Daily Temperature Range	5% DB	MDBR	7.7	10.4	12.6	12.6	13.4	14.3	15.0	14.6	15.1	12.1	11.6	7.7
		MCDBR	10.4	14.8	17.0	17.2	16.4	16.9	17.5	16.4	18.2	16.6	14.6	10.2
		MCWBR	7.0	9.7	9.1	8.8	7.6	6.9	6.5	6.1	8.5	8.7	8.6	6.8
	5% WB	MCDBR	9.0	12.6	14.9	14.7	13.1	13.9	14.3	14.3	15.5	12.5	12.6	8.4
		MCWBR	6.3	8.7	8.1	8.0	6.7	6.3	5.8	6.0	7.8	6.9	7.5	5.8
Clear-Sky Solar Irradiance	taub	0.326	0.353	0.396	0.436	0.413	0.395	0.395	0.399	0.385	0.370	0.338	0.327	
	taud	2.390	2.314	2.202	2.114	2.248	2.335	2.347	2.330	2.361	2.386	2.432	2.424	
	Ebn at Noon	829	854	850	839	867	881	877	864	854	825	810	799	
	Edh at Noon	84	105	131	153	136	125	123	121	110	97	81	76	
All-Sky Solar Radiation	RadAvg	1.71	2.50	3.66	4.87	6.00	6.89	7.26	6.38	4.85	3.28	2.25	1.58	
	RadStd	0.23	0.36	0.35	0.54	0.53	0.63	0.29	0.38	0.39	0.27	0.28	0.23	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (13 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-97	N/A	+46	

Nomenclature: See separate page